MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU–UGP (HONOURS) REGULAR EXAMINATION MARCH 2026
FOURTH SEMESTER
MG4DSCBIF202-DATA STRUCTURES & ALGORITHM
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest() and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

Short Answer Questions
Answer any 5 out of 7 Questions
Each question carries 2 marks

1. Define problem solving in programming. [U][CO1]

2. Recall algorithm? [U][CO1]

3. Define Abstract Data Type (ADT). [U][CO1]

4. What is time complexity? [U][C02]

5. Define stack. [U][CO4]

6. What is a queue? [K][C04]

7. What is linear search? [K][C06]

[2 x 5 = 10]

Part B

Short Essay Questions
Answer any 4 out of 6 Questions
Each question carries 5 marks

8. Describe the characteristics of a good algorithm. [U][CO1]

9. Explain arrays as an ADT with examples. [U][CO1]

10. Summarize singly linked list and its advantages. [U][CO3]

11. Explain stack operations: push and pop. [U][CO4]

12. Explain queue operations: insertion and deletion. [U][CO4]

13. Discuss divide and conquer method with a simple example. [U][C05]

[5 x 4 = 20]

Part C

Long Essay Questions
Answer ALL questions
Each question carries 10 marks

14. Outline time and space complexity of algorithms. Discuss best case and worst case. [U][C02]

15. Explain arrays and linked lists. Compare their advantages and disadvantages. [U][CO3]

16. Describe stack and its applications in detail. [U][C04]

17. Explain sorting techniques: Bubble sort and Selection sort with examples. [U][CO6]

[10 x 2 = 20]

END OF QUESTION PAPER